

Heart Institute of the Midwest

Dept. of Cardiology  
39 Cardiac Way, St. Louis, MO 63101

Discharge Summary

**Patient:** Otto Braun  
**Date of Birth:** 04/18/1946  
**Admission Date:** 09/15/2024  
**Discharge Date:** 09/23/2024  
**Attending Physician:** Prof. Gerhard Thiele, MD

Dear Colleagues,

Mr. Otto Braun, DOB 04/18/1946, was admitted on 09/15/2024 with a 7-day history of progressively worsening dyspnoea on minimal exertion, two-pillow orthopnoea, paroxysmal nocturnal dyspnoea, and bilateral leg swelling. He had gained approximately 6 kg over the preceding two weeks according to his daily weight diary. He has known ischemic cardiomyopathy with a last documented EF of 35%.

**Primary Diagnosis:** Heart failure, not otherwise specified (I50.9)

Lab values on admission:

| Parameter  | Value      | Reference      |
|------------|------------|----------------|
| HbA1c      | 7.9%       | < 7.5%         |
| BNP        | 1240 pg/mL | < 100 pg/mL    |
| Creatinine | 1.6 mg/dL  | 0.7–1.2 mg/dL  |
| Sodium     | 131 mmol/L | 135–145 mmol/L |

Past Medical History: Type 2 diabetes diagnosed 2003, on insulin degludec 24 units at bedtime and acarbose 100 mg three times daily. Chronic atrial fibrillation since 2018, on apixaban 5 mg twice daily. Coronary artery disease, CABG 2014. Hypertension on carvedilol 25 mg twice daily and spironolactone 25 mg. Stage 3a CKD, baseline creatinine 1.4 mg/dL. Ex-smoker, 25 pack-years, quit 2000. Retired factory worker, lives with his daughter who is his primary carer.

Physical Examination on Admission: The patient appeared breathless at rest, unable to complete full sentences. BP 108/72 mmHg, HR 98 and irregular, RR 22/min, SpO2 88% on room air improving to 96% on 4 L/min nasal cannula. Weight 94 kg against dry weight of approximately 88 kg. JVP elevated. Bilateral crackles to mid-zones. No added heart sounds. Moderate hepatomegaly. 3+ pitting oedema to knees.

Investigations: ECG: atrial fibrillation at 98 bpm. BNP 1240 pg/mL, troponin negative. Chest X-ray: bilateral perihilar haziness, upper lobe venous diversion, small bilateral pleural effusions. Echo: EF 28%, dilated LV, moderate functional mitral regurgitation, no new wall motion abnormalities.

Hospital Course: IV furosemide 80 mg twice daily — 4 kg of fluid removed over 4 days. Carvedilol temporarily dose-reduced. IV ferric carboxymaltose for iron-deficiency anaemia (ferritin 18 ng/mL). Dapagliflozin 10 mg added. Insulin degludec reduced from 24 to 18 units given improved oral intake.

Heart Failure Nurse Input: The HF specialist nurse reviewed Mr. Braun on days 4, 6, and 8. Detailed education was provided to both the patient and his daughter on daily weight monitoring, fluid restriction to 1.5 L/day, diuretic dose adjustment protocol (increase furosemide by 20 mg if weight rises more than 2 kg in 48 hours and contact the heart failure hotline), and warning signs requiring emergency attendance. A heart failure action plan card was completed. The daughter confirmed her understanding and the patient was given the 24-hour heart failure helpline number.

Medication at Discharge: Carvedilol 12.5 mg twice daily (to be uptitrated by GP), spironolactone 50 mg, furosemide 80 mg, apixaban 5 mg twice daily, dapagliflozin 10 mg, insulin degludec 18 units at bedtime, acarbose 100 mg three times daily, atorvastatin 40 mg.

Follow-up: Heart failure clinic in 2 weeks for weight, volume, and electrolytes. Cardiology in 6 weeks. Repeat echo in 3 months. GP in 1 week.

### **Intensive Monitoring During Diuresis Phase:**

Mr. Braun was monitored intensively throughout the diuresis phase. Daily weight measurement, strict fluid balance recording, twice-daily vital signs, continuous ECG monitoring, and daily electrolyte and renal function assessment were performed. Potassium supplementation was required on days 2 and 3 in response to diuretic-induced hypokalaemia (K<sup>+</sup> nadir 3.1 mmol/L on day 2). Renal function showed a modest transient decline with creatinine peaking at 1.9 mg/dL on day 3 before improving to 1.5 mg/dL by day 6 as fluid status normalised.

### **Heart Failure Self-Management Education:**

The heart failure specialist nurse reviewed Mr. Braun on days 4, 6, and 8. His daughter attended all three sessions. Education covered: the chronic progressive nature of heart failure and the critical importance of medication adherence; daily weight monitoring (same time each morning after voiding and before eating); fluid restriction to 1.5 litres per day; the diuretic dose adjustment protocol — increase furosemide by 20 mg if weight rises more than 2 kg over 48 hours and telephone the heart failure hotline; recognition of warning signs requiring emergency attendance; dietary salt restriction; and the planned uptitration schedule for carvedilol.

A written heart failure action plan was completed, laminated, and attached to the patient's home medication record. The daughter confirmed full understanding. The patient was provided with the 24-hour heart failure telephone helpline number and instructed to use this service before presenting to the emergency department for non-emergency heart failure symptoms. Sodium intake monitoring diary was started during the admission and continued at discharge.

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Prof. Gerhard Thiele, MD

Director, Heart Institute